

Seminar 9[†]

Statistics 7 (Forecasting) Solutions

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December 13, 2025

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Exercise 1

Exponential Forecasting with $\alpha = 0.2$:

Month	x_t	F_t	$e_t = x_t - F_t$
1	23		
2	39	40.00	
3	36		
4	26		
5	36		
6	12		
7	26		
8	30		
9	42		
10	35		
11	36		
12	28		

Exercise 1 (Cont'd)

1. Complete the above table for exponential forecasting assuming $\alpha = 0.2$.

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Month	x_t	F_t	$e_t = x_t - F_t$
1	23		
2	39	40.00	-1.00
3	36	39.80	-3.80
4	26	39.04	-13.04
5	36	36.43	-0.43
6	12	36.35	-24.35
7	26	31.48	-5.48
8	30	30.38	-0.38
9	42	30.30	11.70
10	35	32.64	2.36
11	36	33.12	2.88
12	28	33.69	-5.69

Exercise 1 (Cont'd)

2. State assumption(s) required for exponential forecasting.

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The time series should be absent of trend and seasonal effect

Exercise 2

Month	x_t	F_t	$e_t = x_t - F_t$
1	45		
2	62	67.00	
3	73		
4	87		
5	86		
6	95		

Exercise 2 (Cont'd)

1. Complete the above table for exponential forecasting assuming $\alpha = 0.2$ and $\alpha = 0.4$.

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When $\alpha = 0.2$, we have:

Month	x_t	F_t	$e_t = x_t - F_t$
1	45		
2	62	67.00	-5.00
3	73	66.00	7.00
4	87	67.40	19.60
5	86	71.32	14.68
6	95	74.26	20.74

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Month	x_t	F_t	$e_t = x_t - F_t$
1	45		
2	62	67.00	-5.00
3	73	65.00	8.00
4	87	68.20	18.80
5	86	75.72	10.28
6	95	79.83	15.17

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MSE when $\alpha = 0.2$:

$$\frac{(-5)^2 + 7^2 + (19.6)^2 + (14.68)^2 + (20.74)^2}{5} = \frac{4492.22}{5} = 898.44$$

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MSE when $\alpha = 0.4$:

$$\frac{(-5)^2 + 8^2 + (18.8)^2 + (10.28)^2 + (15.17)^2}{5} = \frac{3277.33}{5} = 655.47$$

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$$\frac{|-5| + |7| + |19.6| + |14.68| + |20.74|}{5} = \frac{67.02}{5} = 13.40$$

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MAE when $\alpha = 0.2$:

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MAE when $\alpha = 0.4$:

$$\frac{|-5| + |8| + |18.8| + |10.28| + |15.17|}{5} = \frac{57.25}{5} = 11.45$$

Exercise 2 (Cont'd)

4. Based on (2) and (3), suggest which α is more suitable for exponential forecasting.

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Both MSE and MAE are lower when $\alpha = 0.4$. Therefore, $\alpha = 0.4$ is a better choice for exponential forecasting.

Exercise 3

Period	x_t	Period	x_t	Period	x_t	Period	x_t
1	22	8	20	15	22	22	22
2	18	9	25	16	24	23	28
3	15	10	21	17	29	24	31
4	19	11	20	18	27		
5	23	12	20	19	21		
6	21	13	27	20	26		
7	16	14	25	21	29		

Exercise 3 (cont'd)

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1. Compute 3-month simple moving average for the above data.
2. Compute 4-month simple moving average for the above data.
3. Compute centered 4-month simple moving average for the above data.